Lifting & Jacking Equipment

Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Lifting and jacking equipment

What it is used for

Lifting and jacking equipment raises and supports vehicles or heavy components for inspection, repair, and maintenance. Correct equipment selection, rated load limits, approved lifting points, and secure support are essential to prevent crushing injuries and equipment damage.

Main parts

- Vehicle lift or jack raises the vehicle or component.
- Hydraulic pump or screw mechanism generates lifting force.
- Lifting arms, pads, or saddle transfer force to approved lifting points.
- Axle stands or support stands hold the raised load where required.
- Release valve or lowering control lowers the load in a controlled manner.
- Wheel chocks prevent unintended vehicle movement.
- Locking mechanism or safety latch helps secure a raised lift.
- Control switch or handle operates the equipment.

Before you start

1. Only trained operators should use lifting and jacking equipment.
2. Identify the vehicle or load weight and select equipment with a suitable rated capacity.
3. Inspect the jack, lift, arms, pads, saddle, hoses, pins, locks, and stands for damage or leakage.
4. Check the floor is firm, level, and suitable for the equipment and load.
5. Locate the manufacturer's approved lifting points and check the vehicle load distribution.
6. Park the vehicle securely, apply the parking brake, select Park or the appropriate gear, and chock the wheels as required.
7. Keep people, tools, and loose items away from the lifting area.

How to use

1. Position the jack, lift, or lifting arms at the approved lifting points.
2. Confirm that pads or saddles contact the intended points securely and are not resting on weak or damaged components.
3. Raise the load slightly and pause to check balance, contact, and stability.
4. Continue lifting slowly while observing the load and equipment for movement, tilting, or unusual sounds.
5. Engage the lift's mechanical locks where provided and follow the manufacturer's instructions.
6. For jacking equipment, lower the load onto correctly rated axle stands at approved support points before working underneath.

7. Verify the load is stable and the supports are correctly positioned before accessing the underside.
8. Keep the lifting area clear and never move the equipment with a raised load unless the equipment is specifically designed for it.
9. When work is complete, remove tools and people from the danger zone and raise the load slightly if needed to release support stands safely.
10. Lower the load slowly using the designated release control and ensure the area is clear.
11. Remove the equipment, store it securely, and report any defects or unusual operation.

⚠ Safety

- Never work under a vehicle supported only by a hydraulic jack or lifting mechanism without the required mechanical supports.
- Never exceed the rated capacity of the jack, lift, stands, or lifting accessories.
- Do not lift a vehicle at unapproved points or on unstable, sloping, or soft ground.
- Keep hands and feet away from pinch points and never place any part of the body under an unsecured load.
- Do not bypass safety locks, overload protection, or interlocks.
- Stop immediately if the load shifts, the equipment leaks, or stability becomes uncertain; clear the area and follow the emergency procedure.

Care & storage

- Clean lifting pads, saddles, and contact points after use.
- Inspect hydraulic hoses, cylinders, seals, wheels, pins, locks, and stands regularly.
- Check hydraulic fluid and lubricate designated points according to the maintenance schedule.
- Keep equipment dry and store it in the designated location with moving parts lowered safely.
- Arrange authorised inspection and repair for leaks, deformation, damaged threads, or unreliable locking mechanisms.

Common mistakes to avoid

- Using a jack or lift with insufficient rated capacity.
- Positioning lifting pads away from the manufacturer's approved points.
- Working underneath a load without correctly rated mechanical support stands.
- Lowering the load while people, tools, or other obstacles remain in the danger zone.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.